

SOLVED- KTU MODEL QUESTION PAPER

1. In interaction design, user research is the process of gathering and analyzing information about users to understand their needs, behaviors, and attitudes. The goal is to use this information to design better products and services.

User research is a crucial part of the design process because it helps designers bridge the gap between what they think users need and what users actually need. It can be conducted at all stages of the design process, including before designing, during development, and after the product is released.

2. **Satisficing (Detailed explanation with example give 3 marks)**

This is good enough. I don't want to spend more time learning to do it better."

Satisficing is a decision-making strategy where users select the first option that meets their needs sufficiently, rather than searching for the optimal or perfect choice.

User Behavior: Users often choose the first option that seems good enough rather than exploring all possibilities.

Don't Make Me Think , people don't like to think any more than they have to it's work!

3. **Definition 1 mark + importance 2 marks)**

Information architecture (IA) is the art of organizing an information space. It encompasses many things: presenting, searching, browsing, labeling, categorizing, sorting, manipulating, and strategically hiding information.

4. **(Explanation or code give 3 marks)**

List groups are a flexible and powerful component for displaying a series of content. Modify and extend them to support just about any content within.

The most basic list group is an unordered list with list items:

To create a basic list group, use an `<ul>` element with class `.list-group`, and `<li>` elements with class `.list-group-item`:

```
<ul class="list-group">
```

```
<li class="list-group-item">First item</li>
```

```
<li class="list-group-item">Second item</li>
```

```
<li class="list-group-item">Third item</li>
```

```
</ul>
```

5. **Clear entry points (What, when, how and why)**

Present only a few main entry points into the interface; make them task-oriented and descriptive.

Use when

You're designing a site or app that has a lot of first-time or infrequent users.

Most of these users would be best served by reading a certain piece of introductory text, doing an initial task, or choosing from a very small number of frequently used options.

If the purpose is clear to nearly everyone who starts using it, and if adding another navigation step might annoy most users (especially those who are intermediate to expert), this design choice might not be the best.

6. Navbar

A "navbar" is an area on a page that contains navigation components (links, buttons, etc) for getting to other pages of the website.

With Bootstrap, a navigation bar can extend or collapse, depending on the screen size.

A standard navigation bar is created with the .navbar class, followed by a responsive collapsing class: .navbar-expand-xxl|xl|lg|md|sm.

To add links inside the navbar, use either an element (or a <div>) with class="navbar-nav". Then add elements with a .nav-item class followed by an <a> element with a .nav-link class:

Add the .active class on .nav-link to indicate the current page.

Vertical Navbar & Centered Navbar

Colored Navbar

Brand / Logo

Image and text

Fixed Navigation Bar

Sticky Navbar

Scrolling

7. Grid of Equals (What, when, how and why)

Arrange content items in a grid or matrix. Each item should follow a common template, and each item's visual weight should be similar. Link to jump pages as necessary

8. **Module Tabs** in the context of user interface (UI) design refer to a tabbed navigation pattern where different "modules" or sections of content are grouped together under separate tabs within the same interface. This approach allows users to navigate between different views or content areas within the same page or interface, making it easy to switch between them without having to load new pages.

Tabs are typically used to organize related information or features, enabling users to interact with them efficiently without overwhelming the interface with excessive content. Module tabs can be used in many scenarios, such as within dashboards, settings pages, or complex forms, where the content can be logically divided into discrete modules.

9. **Responsive design** is a design approach in web development that ensures a website or application automatically adjusts its layout, content, and behavior to fit a variety of screen sizes and devices. The goal is to create a seamless user experience regardless of the device being used, whether it's a smartphone, tablet, laptop, or desktop computer.

Responsive design uses flexible grid layouts, images, and CSS media queries to adapt the interface based on the screen size, resolution, and orientation of the device. Instead of designing multiple versions of the site for different devices, a single codebase is created, and the layout and content dynamically change to fit the device.

10. Form Controls

- Form controls are the basic building blocks for creating forms in Bootstrap 5. They include input fields, text areas, select menus, and more.

Input Fields:

- Text Input: `<input type="text" class="form-control" placeholder="Enter text">`

Standard text input field with customizable size and appearance.

- Password Input: `<input type="password" class="form-control" placeholder="Enter password">`

Input field for entering sensitive information. Masked by default.

- Email Input: `<input type="email" class="form-control" placeholder="Enter email">`

Validates email addresses and provides built-in browser validation.

- Number Input: `<input type="number" class="form-control" placeholder="Enter number">`

Allows numeric input with optional step and min/max attributes.

- Text Areas:

Textarea: `<textarea class="form-control" rows="3" placeholder="Enter text"></textarea>`

- Multi-line input for longer text entries. Adjustable in size using the rows attribute.

- Select Menus:

```
<select class="form-select" aria-label="Select option">
```

```
<option selected>Select option</option>
```

```
<option value="1">Option 1</option>
```

```
<option value="2">Option 2</option>
```

```
</select>
```

- Drop-down menu for selecting one or more options from a list.

Form Controls Design Considerations:

- Sizing: Use .form-control-lg or .form-control-sm classes to adjust the size of form controls.
- Validation States: Use .is-valid or .is-invalid classes to indicate validation status.

Disabled State: Add the disabled attribute to make a form control non-interactive.

11 a.

The Significance of User Research in Interaction Design

UIST stands for User Interface Software and Technology. It focuses on designing and improving the ways users interact with computers, smartphones, and other devices. This involves making software and hardware easy and enjoyable to use.

- Good interface design doesn't start with pictures.
- It starts with an understanding of people: what they're like, why they use a given piece of software, and how they might interact with it.
- The more you know about them, and the more you empathize with them, the more effectively you can design for them.
- Each time someone uses an application, or any digital product, he carries on a conversation with the machine.
- It may be literal, as with a command line or phone menu, or tacit, like the "conversation" an artist has with her paints and canvas the give and take between the craftsman and the thing being built.

- With social software, it may even be a conversation by proxy. Whatever the case, the user interface mediates that conversation, helping users achieve whatever ends they had in mind
- As a user interface designer, you need to script the conversation, or at least define its terms.
- And if you're going to script a conversation, you should understand the human's side as well as possible.
- What are the user's motives and intentions?
- What "vocabulary" of words, icons, and gestures does the user expect to employ?
- How can the application set expectations appropriately for the user?
- How do the user and the machine finally end up communicating meaning to each other?

There's a maxim in the field of interface design: "Know thy users, for they are not you!"

- **User research aims to understand users' needs, behaviors, and challenges to design better interfaces and interactions.**
- To get a design started, you'll need to characterize the kinds of people who will be using your design, and the best way to do that is to go out and meet them.
- Each user group is unique.
- The target audience for, say, a new mobile phone app will differ dramatically from the target audience for a piece of scientific software.
- Even if the same person uses both, his expectations for each are different a researcher using scientific software might tolerate a less-polished interface in exchange for high functionality, whereas that same person may stop using the mobile app if he finds its UI to be too hard to use after a few days.

The Basics of User Research-METHODS

Direct observation

- Interviews and onsite user visits put you directly into the user's world. You can ask users about what their goals are and what tasks they typically do.

- Interviews can be structured with a predefined set of questions or unstructured, where you probe whatever subject comes up. Interviews gives a lot of flexibility;

Case studies

- They provide deep and detailed views of representative users or user groups.
- Useful for exploring extreme users who push software boundaries, especially in redesign projects.
- Essential for designing custom software for a specific user or site, understanding their context of use is crucial.

Surveys

- Written surveys can collect information from many users.
- Careful survey design is essential. If you want reliable numbers instead of a qualitative “feel” for the target audience, you absolutely must write the questions correctly, pick the survey recipients correctly, and analyze the answers correctly—and that’s a science.

Personas

- Personas aren’t a data-gathering method, but they help to figure out what to do with the given data.
- This is a design technique that “models” the target audiences.
- For each major user group, it create a character that captures the most important aspects of the users in that group:
- What tasks they’re trying to accomplish, their ultimate goals, and their experience levels in the subject domain and with computers in general.
- Personas keep your design focused. As you work, ask yourself, "Would this persona really do X? What would they do instead?"

11b. I want to accomplish something now, not later.

Instant gratification in UIST refers to the user’s desire for immediate rewards or feedback when interacting with a system or interface.

- If someone starts using an application and gets a “success experience” within the first few seconds, that’s gratifying! He’ll be more likely to keep using it, even if it gets harder later.
- They will feel more confident in the application, and more confident in themselves, than if it had taken a while to figure things out.

Importance:

- **User Satisfaction:** Providing quick responses and results enhances the user experience and keeps users engaged.
- **Efficiency:** Users prefer interfaces that offer immediate feedback, which helps them feel productive and reduces frustration.

Design Strategies:

- **Responsive Feedback:** Ensure that the interface provides immediate visual or auditory feedback for user actions.
- **Fast Loading Times:** Optimize performance to minimize delays in processing and displaying results.
- **Streamlined Processes:** Simplify tasks to reduce waiting times and provide instant outcomes. Avoid placing introductory functions behind barriers like registrations, long instructions, slow-loading screens, or advertisements.

Designing for instant gratification makes the experience more enjoyable and user-friendly by giving users quick and satisfying responses.

12 a. Streamlined Repetition

Streamlined repetition involves making repetitive tasks more efficient and less tedious by simplifying and automating them.

Importance:

- **Increases Efficiency:** Reduces the time and effort required to perform repeated actions.
- **Enhances User Experience:** Makes interactions smoother and less frustrating for users.

Design Strategies:

- **Automate Tasks:** Implement features like shortcuts, macros, or automation tools to handle repetitive actions.

- **Consistent Design:** Use consistent layouts and controls to make repeated tasks easier to perform.
- **Provide Feedback:** Offer clear feedback for repeated actions to confirm that tasks are completed successfully.

Designing for streamlined repetition helps users perform repetitive tasks more quickly and with less effort, improving overall efficiency and satisfaction.

EXAMPLE

Find and Replace dialog boxes, commonly used in text editors (like Word or email composers), are designed to simplify repetitive tasks. Users enter the old phrase and the new phrase, then use a single Replace button to change each occurrence one by one. If users are sure they want to replace every instance, they can click the Replace All button, which performs the entire task with just one click.

12 b Habituation

- That gesture works everywhere else; why doesn't it work here, too?"
- Habituation refers to the process where users become familiar with certain actions or gestures in a user interface through repeated use.
- **Benefits:**
- **Expertise:** Users become more skilled and efficient with the tool.
- **Flow:** Helps users work smoothly without needing to think about every action.
- **Risks:**
- **Unexpected Problems:** Users might rely on habitual gestures that don't work in all situations.
- **Confusion:** If a habit doesn't work or causes issues, users may be confused and need to figure out how to fix it or correct mistakes.

Understanding habituation helps in designing interfaces that support efficient use while minimizing potential problems

13 a

Feature search and browse

News steam

Picture manager

Alternative views

Thumbnails

Dashboard

Canvas plus palette

Settings editor

13 b Table

In Bootstrap, tables use the `.table` class for styling.

Since many third-party widgets like calendars use `<table>` elements, Bootstrap's table styles are not applied automatically.

To style a table with Bootstrap, just add the `.table` class to it.

```
<table class="table">
```

Striped Rows & Bordered Table

The `.table-striped` class adds zebra-stripes to a table:

```
<table class="table table-striped">
```

Hover Rows

The `.table-hover` class enables a hover state (grey background on mouse over) on table rows

```
<table class="table table-hover">
```

Black/Dark Table

The `.table-dark` class adds a black background to the table:

```
<table class="table table-dark table-hover"> //combining hover and dark table
```

Borderless Table

The `.table-borderless` class removes borders from the table

14 a

Picture manager (What, when, how and why)

- Use thumbnails, item views, and a browsing interface to create a familiar structure for managing photos, videos, and other pictorial items.
- People use your software to work with lists or collections of pictorial things: photos, drawings, video clips, and so on.

- The list might be in a web page, or in an application, or both.
- It might allow editing by the owner of the content, or it might simply show the content to the public for browsing, viewing, and comments.

14 b

Canvas Plus palette (What, when, how and why)

Place an iconic palette next to a blank canvas; the user clicks on the palette buttons to create objects on the canvas.

This pair of panels

- ☐ a palette for choosing tools
- ☐ a canvas for creating

are common in desktop software.

- They mirror real-life objects, making them easy to understand.
- The palette uses familiar icons, like paintbrushes and magnifying glasses, which are consistently used across different applications.

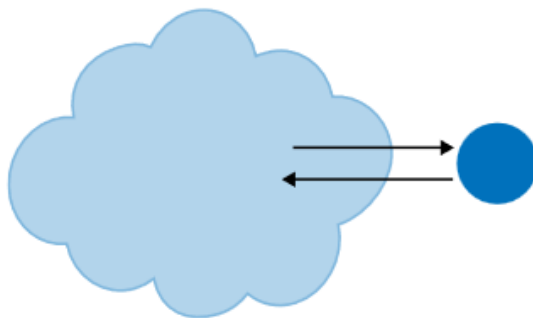
15 a, NAVIGATION MODELS

- Hub and spoke
- Fully connected
- Multi-level
- Stepwise
- Fully connected
- Pyramid
- Pan and zoom
- Flat navigation
- Modal panel
- Clear entry points

- Escape hatch

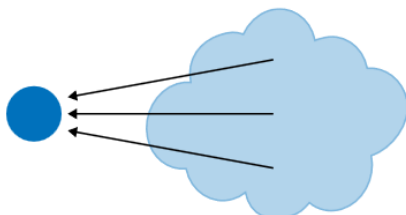
15 b

- This brings a user to a screen with no navigation options other than acknowledging its message, completing its form, or clicking the panel away
- A modal window creates a mode that disables user interaction with the main window but keeps it visible, with the modal window as a child window in front of it. Users *must* interact with the modal window before they can return to the parent window.
- User interfaces typically use modal windows to command user awareness and to display emergency states



16 a.

- When a user is hopelessly entangled in an app, reaches an error state, or gets deep-linked into a page that he has no context for understanding, he needs an escape hatch a well-labeled link to get back to a known place.
- Add detailed explanation of escape hatch



16 b. Effectiveness of Sign-In Tools in Interface Navigation Design

Sign-in tools are an essential part of user interface (UI) design, playing a crucial role in **user authentication** and contributing significantly to the overall **user experience (UX)**. A well-designed sign-in process ensures that users can securely access their accounts while minimizing friction, promoting engagement, and enhancing the overall usability of the application or website.

In today's digital landscape, sign-in tools are not just about verifying user identity but also about offering a seamless, efficient, and secure experience that builds trust and encourages users to engage with a product over time.

Key Elements of Sign-In Tools in Interface Navigation Design

1. Ease of Use:

- **Simplicity:** The sign-in process should be simple, requiring minimal steps. Over-complicating the process with too many fields or unnecessary verifications can deter users.
- **Pre-filled Information:** If the user has previously signed in or registered, pre-filling fields (such as email addresses) can reduce friction and make the process quicker.
- **Clear Call-to-Actions (CTAs):** Buttons like "Sign In" or "Log In" should be clearly labeled and placed in prominent positions, making it intuitive for users to know where to click.

2. Security:

- **Two-Factor Authentication (2FA):** For sensitive applications or services, sign-in tools can incorporate additional layers of security, such as two-factor authentication. While it introduces an extra step for users, it helps protect their accounts and builds trust.
- **Secure Password Handling:** Secure password inputs (with options for showing/hiding the password) and ensuring passwords are encrypted during transmission are important to maintaining user trust.
- **Single Sign-On (SSO):** SSO allows users to authenticate once and gain access to multiple services or platforms without needing to sign in repeatedly. This reduces login fatigue, simplifies authentication, and improves the overall UX.

3. Social Login Options:

- Offering social sign-in options (such as "Sign in with Google" or "Sign in with Facebook") can significantly streamline the authentication process. Social logins allow users to bypass creating and remembering new usernames and passwords, which increases conversion rates for sign-ups and logins.
- **Pros of Social Sign-Ins:**
 - Faster sign-in process, especially for users who don't want to create another account.
 - Reduced friction, as users may already be logged into their social accounts.
- **Considerations:**
 - Some users may be hesitant to link their social media accounts to external apps due to privacy concerns. Offering alternative sign-in

options (such as email/password) alongside social logins can address this concern.

4. Password Recovery/Reset:

- A seamless **password recovery** tool is crucial in maintaining a smooth UX. This feature allows users to quickly regain access to their accounts without unnecessary barriers, such as long wait times or confusing processes.
- Providing a “Forgot Password” link with simple instructions for recovery or reset can reduce user frustration and prevent abandonment.

5. Progressive Disclosure:

17 a.

Definition of page layout

Page layout is the art of manipulating the user’s attention on a page to convey meaning, sequence, and points of interaction

- several elements of page layout:
 - **visual hierarchy**
 - **visual flow**
 - **how to use dynamic displays.**

Layout patterns

Centerstage

grid of equals

module tabs

accordion

collapsible panels

diagonal balance

liquid layout

17 b

Stacks modules of content vertically.

Allows independent opening/closing of each module.

Ideal for heterogeneous content such as text blocks, lists, images, etc.

User-friendly for limited space pages

In UI (User Interface) design, accordions are a popular interactive component used to manage and organize large amounts of content within a limited space. They are essentially collapsible sections or panels that expand when clicked to reveal more content, and collapse again when clicked to hide the content. The purpose of accordions is to enhance the user experience by enabling efficient navigation through multiple sections of content without overwhelming the screen or requiring excessive scrolling.

Key Features of Accordions:

Expandable and Collapsible Sections: Users can expand or collapse individual sections of content.

Space Efficiency: They allow users to view content in a compact area, reducing clutter.

Smooth Interaction: Typically, accordions feature smooth animations that make expanding and collapsing sections feel intuitive.

Single or Multiple Open Sections: Accordions can be designed to allow only one section to be open at a time (accordion style), or multiple sections can remain open simultaneously.

18 a.Diagonal balance

Page elements are in an asymmetric fashion, but balance it by using visual weight

- For a Layout, a logo or header is at the top left, while a button or interactive link at the bottom right

Diagonal balance in design refers to the use of diagonal lines or elements to create a sense of equilibrium and visual harmony in a composition. This principle is especially important in visual design (graphic design, interior design, etc.) because it can introduce a dynamic and energized feeling, leading the viewer's eye across the design in an organized manner.

Key Concepts of Diagonal Balance:

- **Dynamic Composition:** Diagonal lines create a sense of movement and energy, guiding the viewer's eye through the design. This can add excitement, flow, and visual interest to an otherwise static layout.
- **Asymmetry:** Unlike traditional symmetrical balance (where elements on either side of the center line are identical), diagonal balance often involves asymmetry. Elements placed diagonally across the design can feel balanced, even if they differ in size or weight, as long as they visually counterbalance each other.
- **Flow and Direction:** Diagonal balance often involves creating a visual path that guides the viewer's eye from one area of the design to another. The direction of the diagonal line (from top left to bottom right, or top right to bottom left) can affect how the viewer navigates the design.
- **Visual Tension:** Diagonal lines can create a sense of tension or action. This tension is often balanced with counteracting elements such as size, color, or weight to keep the overall composition harmonious and not overwhelming.
-

18 b.Liquid layout

A **liquid layout** (also known as a **fluid layout**) is a web design approach where the elements of a webpage adjust dynamically to fit the width of the browser window. Unlike fixed layouts, where the width of the content is defined in pixels, a liquid layout uses percentages or relative units like em or vw (viewport width) to set the size of elements, allowing them to resize based on the user's screen or browser window size.

Key Features of Liquid Layout:

1. **Responsive to Screen Size:** The layout stretches or shrinks proportionally based on the width of the browser window or device screen, making it well-suited for responsive design.
2. **Fluid Grids:** Instead of specifying exact pixel values, liquid layouts use percentage-based values for widths, margins, and padding. This allows elements to resize smoothly when the browser window is resized.
3. **Flexible Content:** The text, images, and other content adjust in size, ensuring that the page looks good on any device, from desktop to mobile.

19 a

Progress Indicators

- **Purpose:** Progress indicators show the status of ongoing processes, such as loading, file uploads, or form submissions. They provide feedback on how much of a task is completed and how much remains, helping to manage user expectations and reduce frustration.

Types of Progress Indicators:

- **Progress Bars:** Horizontal bars that fill up as the task progresses, showing a percentage or visual representation of completion. They can be determinate (showing exact progress) or indeterminate (showing a general sense of activity, often with an animated element).
- **Spinners/Loading Indicators:** Animated elements (e.g., spinning circles) that indicate that a process is in progress. They are typically used when the duration of the task is unknown or variable.
- **Percentage Indicators:** Display the exact percentage of completion, often used in conjunction with progress bars for more precise feedback.
- **Step Indicators:** Show progress through a multi-step process, often used in wizards or onboarding flows. They indicate which step the user is currently on and how many steps are remaining.

Design Considerations:

- **Clarity:** Make sure progress indicators are easily recognizable and provide clear feedback on the current status of the task. Avoid ambiguity about whether the process is ongoing or stalled.
- **Feedback:** Use different types of progress indicators appropriately based on the nature of the task. For example, use spinners for tasks with unknown duration and progress bars for tasks with a clear endpoint.
- **Visibility:** Ensure progress indicators are placed where users can easily see them without disrupting their workflow. They should be prominent enough to draw attention but not overly distracting.
- **Performance:** For tasks that take a significant amount of time, consider providing additional information or options (like cancel buttons) to manage long-running processes effectively.

19 b Hover Tools

- **Purpose:** Hover tools, or hover effects, provide additional information or interactive elements when the user places their cursor over a UI element. They enhance usability by offering context or options without cluttering the interface.

Types of Hover Tools:

- **Tooltips:** Small pop-ups that appear when the user hovers over an element, providing brief explanatory text or additional information.
- **Hover Menus:** Menus that appear when hovering over a specific item, revealing additional options or actions related to that item.
- **Hover Effects:** Visual changes such as color shifts, enlargements, or animations that provide feedback and make elements appear interactive.

Design Considerations:

- **Visibility:** Ensure hover tools appear consistently and are easy to spot. Tooltips should be clearly visible and not obstruct other important content.
- **Timing:** Avoid delays or overly long hover effects that could frustrate users. Hover tools should appear and disappear quickly and smoothly.
- **Accessibility:** Provide equivalent functionality for users who do not use a mouse, such as touchscreens or keyboard navigation, and ensure hover effects are not the only way to access important information.

20 a

Implementing **multi-level undo functionality** effectively while maintaining **interface consistency** and **performance** is a crucial challenge for designers and developers. Users expect the ability to reverse actions step by step, but this functionality needs to be implemented in a way that doesn't overwhelm the interface or degrade performance, especially in applications with complex user interactions or large datasets.

Here are several key considerations and best practices for designing and implementing multi-level undo functionality effectively:

1. Clear Undo/Redo Controls

- **Consistency in Placement:** Place undo/redo buttons in predictable locations to ensure easy access. Common places are in the toolbar, context menus, or via keyboard shortcuts (Ctrl+Z for undo, Ctrl+Y for redo).
- **Visual Feedback:** Provide visual cues that inform users of the available actions. This includes showing the availability of undo/redo options (e.g., graying out the button if no actions can be undone).
- **Iconography:** Use intuitive icons, like backward arrows for undo and forward arrows for redo, to make these options easily identifiable and accessible.

2. User-Friendly History Navigation

- **History List:** Show the history of actions in an accessible way (e.g., a timeline or list). For example, in graphic design tools like Photoshop, the history panel shows a sequence of actions, allowing users to jump to a specific state.
- **Grouped Actions:** Group related actions (e.g., text editing, object manipulation) together in the undo history to prevent users from being overwhelmed by the number of individual steps.
- **Label Actions:** For long or complex actions, label the undo steps clearly (e.g., “Deleted Text” or “Moved Object”) to help users understand what each step represents.

20 b

Designing and implementing **macros** in user interfaces to support workflows and task automation is an important feature that can significantly improve efficiency and productivity. Macros allow users to automate repetitive tasks or sequences of actions, and they can be implemented in various applications like spreadsheets, graphic design software, and development environments.

Here are some best practices for designing and implementing macros in interfaces to ensure they are both **user-friendly** and **effective**:

1. Clear Macro Definition and Recording

- **Simple Recording Interface:** Provide an intuitive interface for recording macros. The user should be able to easily start and stop recording, and the interface should automatically capture relevant user actions (e.g., keystrokes, clicks, drag-and-drop actions). This can be achieved through a button like "**Record Macro**" or a hotkey.
- **Preview and Edit Recorded Macros:** After recording, users should have an option to review and edit the sequence of actions. This could include:
 - **Timeline View:** A timeline or step-by-step list showing each action performed during the recording process.
 - **Editable Scripts:** Users should be able to modify the recorded actions, either through a graphical interface or by directly editing the underlying script/code (for advanced users).
- **Action Grouping:** Organize related actions into groups to make the macros more readable and maintainable. For example, all formatting actions in a word processor could be grouped, making it easier for users to understand what the macro does.